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Towards explicit soft skills labelling in ESCO through semantic NLP analysis

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Abstract

The classification and identification of soft skills within the European Skills, Competences, Qualifications, and Occupations (ESCO) framework presents a significant challenge, as ESCO does not explicitly indicate which of its listed skills are considered “soft”. This lack of clear labelling complicates the strategic alignment of educational programs and workforce development initiatives with the evolving demands of today’s labour market, where soft skills are increasingly recognised as essential for professional success. Our study seeks to address this challenge by conducting a systematic literature review to identify and define a comprehensive set of soft skills, and by employing Natural Language Processing (NLP) techniques to map these skills to the ESCO taxonomy. Through semantic analysis and comparison of soft skills identified from scientific literature against the ESCO skillset, we establish a refined classification that explicitly labels soft skills within the ESCO framework. Our findings provide a necessary clarification of soft skills within the standardised European context, offering valuable insights for policymakers, educators, and HR professionals focused on fostering soft skills development. By clarifying the soft skills embedded within the ESCO framework, this research contributes to the advancement of skill taxonomies and promotes a more integrated approach to the development and assessment of soft skills.

JEL classification J20, J24, I26, C88.

Keywords Softs skills, ESCO, NLP, Semantic similarity, Labour market

1 Introduction

In the rapidly evolving landscape of the 21st-century labour market, soft skills - encompassing personal attributes, personality traits, inherent social cues, and communication abilities - have become indispensable components of employment, education, and personal development. Recent advancements in the business environment, driven by technological progress and the

increasing competitiveness and dynamism of the global economy, have contributed to the growing demand for new skill sets (Dolce et al. 2020). Unlike hard skills, which are often measurable and technical in nature, soft skills involve a broad spectrum of interpersonal and cognitive abilities, including teamwork, leadership, creativity, and emotional intelligence. These capabilities are increasingly recognised for their critical role in determining professional success, facilitating smooth integration into diverse work environments, and enhancing individual adaptability in an era characterised by continuing technological advancements and rapidly changing occupational demands (Clarke 2018; Succi 2019; Toner 2011; Tsirkas et al. 2020).

The classification and identification of soft skills within frameworks, such as the European Skills, Competences,

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Qualifications, and Occupations (ESCO), encounter significant challenges due to the absence of a distinct labelling system explicitly dedicated to soft skills. While ESCO, incorporates these competencies under the broader category of “Transversal Knowledge, Skills, and Competences”, this category includes a wide and heterogeneous range of abilities, such as cognitive, digital, and even physical/manual skills, that extend well beyond the traditional conceptual scope of soft skills¹. Although ESCO recognises the foundational role of these transversal competencies in personal development and their importance for acquiring technical skills necessary for labour market integration, the lack of a precise classification for soft skills specifically limits the ability to target, analyse, and apply them effectively in educational design and workforce development strategies. Given the increasing recognition of soft skills as critical factors for professional success, there is a clear need to delineate them more explicitly within existing frameworks. Introducing a more refined soft skills classification allows for improved alignment between education and training initiatives and the evolving demands of the labour market, offering a more operational and context-sensitive tool for stakeholders.

To address these challenges, our research aims to identify which ESCO skills can be classified as soft skills by applying advanced Natural Language Processing (NLP) techniques. Through the extraction of soft skill examples from literature and their systematic alignment with the skills listed in the ESCO framework, this study establishes a refined classification that explicitly identifies and categorises specific ESCO skills as soft. This classification enhances the representation of soft skills within ESCO, differentiating them from other skill types and providing essential insights for aligning educational programs and workforce development with current labour market needs.

The structure of this article is organised as follows: Sect. 2 introduces the ESCO framework, detailing its components, practical applications, and significance in addressing the challenges of workforce development and labour market analysis. Section 3 reviews the literature on the identification and classification of soft skills, examining the various definitions and terminologies used across disciplines. Section 4 outlines the methodological approach used to match soft skills with ESCO-listed skills. In Sect. 5, the results are presented, highlighting both exact and semantic matches between soft skills identified in the literature and those within the ESCO

framework. This section also provides an analysis of the distribution of soft skills within ESCO, with a focus on cross-sector and sector-specific applications. Finally, Sect. 6 concludes with recommendations for refining soft skill classifications within ESCO, alongside potential implications for policy, education, and workforce development.

2 Understanding ESCO: framework, function, and relevance

The European Skills, Competences, Qualifications, and Occupations (ESCO) framework, developed and maintained by the European Commission’s Directorate-General for Employment, Social Affairs and Inclusion (DG EMPL), constitutes a standardised, multilingual taxonomy designed to bridge the gap between labour markets and education systems (European Commission: Directorate-General for Employment 2017). Since the launch of its first pilot version (ESCO v01) in 2015 (European Commission n.d.), the framework has played a crucial role in aligning the education, training, and employment sectors to enhance workforce adaptability, mobility, and career development. The ESCO platform is freely accessible online, and the latest version (ESCO v1.2.0, released in May 2024) offers detailed descriptions of 3,039 occupations and 13,939 skills, all translated into 28 languages, including the official languages of the European Union, as well as Icelandic, Norwegian, Ukrainian, and Arabic (European Commission 2024).

The framework is organised into three interconnected pillars: occupations, skills and competences, and qualifications. The occupations pillar, drawing upon the International Standard Classification of Occupations (ISCO) (International Labour Office 2023), defines comprehensive job profiles that involve similar content in terms of tasks and require comparable types of skills. Each occupation is clarified through detailed descriptions and scope notes, which establish its semantic boundaries and distinguish it from related or alternative occupations. This alignment ensures that labour market demands are effectively matched with the skills and capabilities of individuals.

The skills and competences pillar categorises cognitive, practical, and interpersonal abilities within a hierarchical framework. These abilities are subdivided into four classifications: knowledge, language skills and knowledge, skills, and transversal skills and competences. Notably, the ESCO framework does not differentiate between “skills” and “competences”, instead using a unified approach to their classification.

The qualifications pillar establishes connections between certifications and the relevant skills and occupations, ensuring alignment between individual competencies and labour market demands. Only qualifications

¹ ESCO identifies six main categories of transversal skills and competences: (1) Core skills and competences; (2) Thinking skills and competences; (3) Self-management skills and competences; (4) Social and communication skills and competences; (5) Physical and manual skills and competences; and (6) Life skills and competences.

mapped to European Qualifications Framework (EQF) levels (European Commission: Directorate-General for Employment 2018) through national referencing are included, providing contextual relevance and consistency.

Each concept within ESCO is assigned human-readable names, which are categorised into three types of labels: preferred terms, non-preferred terms, and hidden labels. Preferred terms (prefLabel) serve as the primary descriptors and are always neutral in gender, with one term per language. Non-preferred terms (altLabel) provide alternative labels, including synonyms, spelling variants, abbreviations, and gender-specific grammatical forms. Hidden labels (hiddenLabel) account for common spelling errors or politically sensitive terminology without being displayed to users. Additional annotations, such as scope notes, reusability levels, and links between skills and occupations, ensure clarity, usability, and contextual relevance.

The ESCO framework enhances job-matching processes by functioning as a semantic dictionary for skills, qualifications, and occupations. Its Linked Open Data format facilitates seamless integration into digital systems, enabling applications such as job matching, training recommendations for reskilling and upskilling, skills mapping, career guidance, and self-assessment tools. Numerous digital platforms and career management systems leverage ESCO's taxonomy to provide automated job-role matching and personalised career recommendations, thereby improving the efficiency of job-matching processes (Boselli et al. 2017; Rentzsch and Staneva 2020; Vrang et al. 2014). Moreover, ESCO facilitates the identification of transferable skills across sectors, thereby promoting career mobility and adaptability.

By harmonising the classification of occupations, skills, and qualifications across regions and contexts, the ESCO framework provides a robust foundation for mapping, analysing, and enhancing labour market dynamics. This represents a significant advancement in workforce development and labour market analysis.

2.1 Literature review

The concept of soft skills encompasses a wide range of non-technical abilities that are crucial for personal development, educational achievement, and professional success. Despite their recognised importance, there remains a notable lack of consensus regarding the precise definition of soft skills. Current practices for identifying and cataloging soft skills vary significantly, resulting in notable discrepancies in how these skills are understood, taught, and assessed across contexts. Often contrasted with hard skills, soft skills are more associated with emotional intelligence (EQ) rather than intellectual intelligence (IQ), with Goleman (1996) being one of the earliest to highlight this distinction. Siekmann and Fowler (2017)

and Succi (2019) further observe that, unlike hard skills, soft skills are inherently challenging to categorise due to their subjective nature and the contextual variations in how they are perceived. Over the years, soft skills have been referred to by various terms such as life skills (World Health Organization 1994), social skills, generic competencies, interpersonal skills, leadership skills, transversal skills, and even key competences for lifelong learning (European Union 2006). Furthermore, the discussion around soft skills has increasingly intersected with concepts such as 21st-century skills and skills for the future of work, where digital competencies are also often included (Ananiadou and Claro 2009). This plurality of terminologies has only added to the complexity of establishing a clear and unified understanding of soft skills in both academic and professional contexts. The challenge of defining soft skills has been a persistent issue in academic literature, with scholars presenting diverse and often divergent perspectives on what constitutes these skills. Definitions vary widely, reflecting differing interpretations and emphases across studies. The term “soft” itself is often substituted by words like “generic”, “fundamental”, “key”, “personal”, “transferable”, or “emotional” depending on the context. Similarly, the term “skills” can be ambiguously replaced with “abilities”, “competencies”, or even “qualities”, contributing to further confusion (Bennett et al. 1999; Chamorro-Premuzic et al. 2010; Matteson et al. 2016).

Davis-Blake et al. (2002) offered an early definition of soft skills as “skills and traits pertaining to personality, attitude, and behaviour”, distinguishing these from formal knowledge or technical expertise and broadening the concept beyond technical competencies. Later, Heckman and Kautz (2012) expanded on this foundation by describing soft skills as personality traits, goals, motivations, and preferences valued in the labour market, educational settings, and beyond. They highlighted the “Big Five” personality traits - openness, conscientiousness, extraversion, agreeableness, and emotional stability - as core components, underscoring the role of personality in defining soft skills. Similarly, Yorke (2006) characterises soft skills as a “combination of dispositions, understandings, attributes, and practices”. Adding to this, Knight and Page (2007) refer to them as “tricky competences” due to their elusive and context-dependent nature, highlighting their continuous development throughout one's life. Haselberger et al. (2012) take a more comprehensive view, defining soft skills as the dynamic combination of cognitive, metacognitive, interpersonal, intellectual, and practical skills. These perspectives underscore the complex nature of soft skills and their critical role in enabling individuals to navigate both professional and personal challenges effectively. As highlighted by Bennett et al. (1999), Chamorro-Premuzic et al. (2010), and Dolce et

al. (2020), the definition of soft skills remains elusive, primarily due to semantic differences in how these skills are described across various sectors and languages.

The fluid nature of soft skills also leads to debates about the distinction between soft and hard skills, with some scholars arguing that the line between the two is often blurred. For instance, Papoutsoglou et al. (2017) define soft competences as the “ability to use knowledge, personal, social, and/or methodological skills” in work or study contexts, while hard competences are associated with the practical application of knowledge and problem-solving abilities. González-Eras and Aguilar (2019) suggest that competences can be seen as a demonstration of the application of generic skills in specific contexts. In some cases, what is considered a soft skill in one field, such as singing, may be classified as a hard skill in another, as in the case of professional performers. This interchangeable nature of skills, depending on context, underscores the challenge of establishing universally applicable definitions and classifications (Khaouja et al. 2019).

The complexity of soft skills definition extends to their classification, with various scholars proposing diverse taxonomies to clarify their scope. Cinque and Ciappei (2014) for example developed a taxonomy that divides soft skills into four categories based on the dimension of actions: political, strategic, organisational, and ethical. These categories are further highlighted across personal, social, and methodological dimensions, illustrating the multi-layered nature of soft skills.

As definitions of soft skills have evolved, the demand for rigorous methodologies to effectively identify, classify, and analyse these complex competencies has similarly intensified. Traditional qualitative approaches, including interviews and surveys, initially provided valuable insights into the role and perception of soft skills. However, the growing complexity of these skills, compounded by their varied interpretations across disciplines and industries, has highlighted the limitations of these methods in capturing the full range of soft skills. In response, recent advancements in technology, particularly Natural Language Processing (NLP), have emerged as promising tools to address these challenges, allowing for more systematic and scalable approaches to soft skills analysis and integration into frameworks such as ESCO.

Chiarello et al. (2021) illustrates the potential of advanced NLP techniques through a text mining analysis aimed at evaluating the adaptability of ESCO framework in addressing skill demands driven by Industry 4.0. Through this analysis, they underscore that utilising methods like Named Entity Recognition (NER) and lexical searches can effectively facilitate the integration of emergent technological competencies into ESCO, ensuring its responsiveness to rapid advancements. Further

expanding on the application of NER techniques, Fareri et al. (2021) developed SkillNER, a system trained to extract soft skills from unstructured text sources such as scientific literature and job descriptions within the ESCO framework. Continuing this exploration, Khaouja et al. (2021) reviewed 108 research articles on the topic of identifying skills from online job postings, emphasising that NLP and information extraction techniques have been critical in building customised databases of skills from textual knowledge bases. Machine learning techniques, including neural networks and word embeddings, are also used to capture skills from text-based resources with increasing precision. A notable study by Zhang et al. (2022) applied machine learning techniques using noisily labeled data to train models to classify skills extracted from job advertisements. The researchers demonstrated that their model's performance in identifying skills closely matched human extraction, showing that automated systems could be nearly as effective as human experts in identifying and classifying job-related skills, including soft skills. However, the study also highlighted some persistent challenges, particularly with NLP's ability to fully capture the subtleties of soft skills descriptions. Moreover, many studies still rely on human judgment for coding and categorising soft skills, introducing potential biases and subjectivity into the research process.

While significant progress has been made in understanding and applying technologies like NLP and machine learning to classify and analyse soft skills, substantial gaps remain in the standardisation of definitions, categorisation, and evaluation methodologies. This lack of uniformity complicates the alignment of educational frameworks with labour market needs, impeding the development of cohesive strategies for soft skills education and assessment (Bak and Jordan 2018; Balcar 2016). The absence of a standardised methodology across frameworks such as developed by the OECD, CEDEFOP or the US O*NET system further challenges policymakers, educators, and industry specialists in building a workforce equipped not only with technical proficiency but also essential interpersonal and cognitive abilities (Cinque and Ciappei 2014; Siekmann and Fowler 2017). Initiatives such as ESCO's recent labeling of “green” skills (European Commission 2022) underscore the need for similar efforts to explicitly categorise soft skills, ensuring that these competencies are effectively integrated into educational curricula and workforce development strategies to meet labour market requirements more effectively.

3 Methodology

Building on the challenges involved in defining and classifying soft skills, this research adopts a novel methodological approach by combining a structured literature

review with advanced Natural Language Processing (NLP) techniques. By integrating empirical insights from the literature with the semantic capabilities of NLP, the study systematically identifies and categorises soft skills within the ESCO framework. This approach enhances both the alignment and comprehensiveness of soft skill classification, contributing to a more robust and scalable methodology for soft skill mapping.

In the initial stage, a targeted literature review was conducted to systematically extract and analyse specific soft skills as defined and conceptualised by various scholars. An analysis of the literature, particularly of scientific articles focused on soft skills, is analogous to consulting subject matter experts. However, in this context, scientific articles draw upon a more extensive and diverse pool of expert knowledge, thereby providing a more comprehensive and reliable perspective. Our focus was exclusively on identifying soft skills without synthesising or reclassifying existing definitions. The literature review was conducted using the Web of Science database, selected for its broad coverage of high-impact, peer-reviewed scholarly publications. This selection ensured that the soft skills identified were drawn from credible and relevant sources. The following keywords were employed: “soft skill”, “soft skills”, “soft-skills”, and “soft-skill”. These variations were included to account for different terminologies and hyphenations across publications. The search was restricted to the “Topic” field, which encompasses the article title, abstract, and keywords, to ensure that the articles explicitly referenced or focused on soft skills as a central aspect of their discussion or conceptual framework. Subsequently, a filter was applied to include only review articles, as they typically provide a synthesis of existing research, offering a more complete and consolidated understanding of soft skills concept. This approach enabled us to capture diverse conceptualisations of soft skills, thereby enhancing the robustness of the dataset by incorporating insights from multiple sources rather than relying on isolated empirical studies. After downloading the initial list, we conducted a preliminary analysis of the titles and abstracts to refine the selection. Articles mentioning soft skills only in passing or in unrelated contexts were removed, ensuring that the final list was closely aligned with the research focus. In cases where the relevance of an article was ambiguous, a second, more detailed review of the introduction or conclusion was performed to confirm its suitability. This filtering process ensured that the final list consisted of articles making substantive contributions to the understanding of soft skills, thereby enhancing the precision and relevance of the study. Following the refinement process, 134 articles that met the inclusion criteria and directly addressed soft skills were identified. An additional 62 papers, identified during the documentation process as particularly

relevant to our research objectives, were manually included. These papers were not retrieved through the initial Web of Science search but were identified primarily through citations within the initially selected review articles or encountered during the broader literature review process. Many of these sources consisted of reports, books, or other relevant publications that were not captured due to the limitations of the original search strategy (e.g., keyword scope or database indexing). To ensure a more comprehensive and representative foundation for documenting the typologies and conceptualisations of soft skills, we chose to include these additional references. They were screened for relevance following a procedure similar to that applied to the database-sourced articles. Consequently, the final selection comprised 196 articles. All sources, including those manually added, are listed in Annex (1). Each of these articles was carefully reviewed, with a focus on sections where soft skills were explicitly mentioned, such as in lists, tables, or highlighted textual elements. Explicit mentions of soft skills, as well as implicit references identified through contextual analysis, were manually extracted. Specific keywords, including “skill”, “soft skills”, and “non-technical skills”, along with related and contextual terms such as “interpersonal skills”, “emotional intelligence”, and “professional competencies” were employed to identify relevant terms within the text. The extracted terms were then consolidated by removing duplicates and grouping synonymous concepts, resulting in a refined list of unique soft skills. This systematic and manual methodology ensured accuracy and relevance while maintaining transparency and reproducibility. Both authors independently conducted the extraction process and cross-validated their results to maintain consistency in identifying and categorising soft skills from the literature. Initially, a total of 683 skills were extracted. Following the removal of duplicates, 202 distinct soft skills were identified and retained. To ensure the consistency of our soft skills classification, both authors independently evaluated each item in the compiled skill list, assigning a code of 1 to skills considered “soft” and 0 otherwise. Agreement was reached on most items, with a percentage agreement of 96.43%, reflecting a high degree of consistency in the evaluation process. Only a small number of discrepancies were recorded, for example, “customer care”, “change management”, or “interacting with clients”, which required further discussion due to their context-dependent interpretations. The authors jointly discussed the items on which they disagreed, clarifying definitions and re-examining the skill descriptions and their contextual usage. Considering the substantial initial agreement and the collaborative resolution of divergent cases, the overall reliability of the classification process was considered robust. The complete list of soft skills extracted from literature is presented in

Annex (2) These skills constituted the dataset used for comparison with the ESCO classification.

In the second stage, we sought to identify which skills within the ESCO framework could be categorised as soft skills by matching them with the soft skills extracted from the literature. We utilised the English-language version 1.2.0 of the ESCO framework, released on 13 May 2024 as the basis for our analysis. To construct the dataset of ESCO skills, we extracted both the preferred labels, and the alternative labels provided for each concept in ESCO. These labels formed the ESCO skills dataset, which we then systematically compared with the dataset of soft skills derived from the literature. The ESCO dataset consisted of 13,939 preferred labels and 86,886 alternative labels.

With embeddings generated for both the soft skills extracted from scientific literature and the skills listed in the ESCO framework, the final stage of the analysis involved matching these two sets of skills.

In the initial phase of our matching process, we employed exact string matching to identify identical skills between the soft skills dataset, extracted from the literature and the ESCO dataset. To enhance the robustness of this comparison, we introduced a lemmatisation step during preprocessing, enabled the reduction of words to their base or dictionary forms prior to matching². Each skill phrase was tokenised, and the individual tokens were lemmatised and then recombined into a single, normalised string. This linguistic normalisation, applied to both datasets, addressed variations in word forms (e.g., singular/plural, verb tenses) and improved the likelihood of identifying equivalent expressions during the exact matching process. The comparison was conducted in a case-insensitive manner with punctuation removed and whitespace normalised. The matching process involved systematically comparing each lemmatised skill from the literature-derived dataset with the lemmatised entries in the ESCO dataset using straightforward lookup functions to identify perfect string matches. While exact matching is highly reliable it has limitations, such as the inability to capture synonyms, variations in word forms, or different word orders.

To overcome the limitations inherent in exact matching and to identify a broader range of correspondences between soft skills identified in the literature and those catalogued within the ESCO framework, the subsequent phase involved the application of Natural Language Processing (NLP) techniques. This computational approach facilitated a systematic alignment and cross-referencing of the two datasets beyond identical lexical matches. Semantic similarity analysis was employed to

detect nuanced relationships between skill descriptions, enabling the identification of semantically related skills despite variations in exact wording. Semantic similarity measures the degree of likeness between two pieces of text based on the meaning they convey. Unlike syntactic similarity, which focuses on the structural aspects of language (e.g., grammar, syntax), semantic similarity explores contextual and conceptual parallels. This process is crucial for various applications, including information retrieval, text summarisation, sentiment analysis, and machine translation. Semantic similarity is rooted in the idea that words, phrases, sentences, and even larger text segments can have varying degrees of relatedness based on their meanings. For instance, “car” and “automobile” are semantically very similar, whereas “car” and “tree” are not. The challenge lies in quantifying this similarity in a way that computers can process.

Prior to conducting the semantic analysis, extensive data preprocessing was carried out to standardise and clean both the literature-derived dataset and the ESCO dataset. This process ensured data consistency and improved the accuracy of the matching procedure. Key preprocessing steps included converting all text to lowercase, removing punctuation, special characters, and excess whitespace, while preserving stop words in multi-word expressions to maintain semantic integrity. Additionally, a typo-tolerant correction step was implemented to automatically identify and correct common spelling errors, thereby reducing the likelihood of missing valid soft skills due to typographical noise. Duplicate entries were eliminated to prevent bias in the matching results³.

To quantify the semantic similarity between the skills in the ESCO dataset and the predefined soft skills from the literature, we employed Sentence-BERT (SBERT)⁴, a pre-trained model optimised for generating dense vector embeddings of sentences and phrases. SBERT was chosen for its efficiency in capturing sentence-level semantic similarity across both single- and multi-word expressions (Choi et al. 2021; Reimers and Gurevych 2019; Stankevičius and Lukoševičius 2024). To evaluate the robustness of our embedding model selection strategy, we conducted a comparative analysis of two SBERT variants: “all-MiniLM-L6-v2” and “paraphrase-MiniLM-L6-v2”. Both models are based on the MiniLM architecture but are fine-tuned for distinct objectives (“all-MiniLM-L6-v2” for general-purpose sentence embeddings and “paraphrase-MiniLM-L6-v2” for paraphrase detection and semantic equivalence). Each model was used to compute cosine similarity scores between

² Lemmatisation was performed using the English language model “en_core_web_sm” from the “spaCy” library in Python.

³ Text preprocessing was conducted using native Python functions along with “spaCy” and “TextBlob”.

⁴ The SBERT model was implemented using the “sentence-transformers” library in Python.

literature-derived soft skills and ESCO skill descriptions. While both yielded a comparable number of identified soft skills, “paraphrase-MiniLM-L6-v2” demonstrated better alignment with manually validated soft skills and produced fewer false negatives. Based on this improved semantic precision, “paraphrase-MiniLM-L6-v2” was selected for the final classification pipeline. However, transformer architectures beyond the Sentence-BERT framework were not evaluated in this study. Models such as “all-mpnet-base-v2” (based on MPNet), “distil-RoBERTa-base”, and “e5-base” may present alternative trade-offs in terms of accuracy, semantic generalisation, and computational efficiency. Future research could extend this analysis by benchmarking a wider range of transformer-based models to enhance the robustness and granularity of soft skill identification across varied textual formulations.

The following mathematical representation outlines the computational methodology employed to match soft skills from literature with ESCO skills using the selected embedding model.

Let $T_{lit} = \{t_1, t_2, t_3 \dots t_n\}$ represent the set of skill terms from the literature dataset and $T_{esco} = \{s_1, s_2, s_3 \dots s_m\}$ represent the set of skills from the ESCO dataset. In both datasets terms were represented by a single word or a multi word expression (e.g., “communication” or “manage other people’s feelings”). Each skill description t_i from the literature dataset and s_j from the ESCO dataset was encoded as a vector embedding using the SBERT model M :

$$v_i^{lit} = M(t_i), v_j^{esco} = M(s_j) \quad (1)$$

where $v_i^{lit} \in \mathbb{R}^d$ and $v_j^{esco} \in \mathbb{R}^d$ and d is the dimensionality of the embedding space. This process generated two sets of embeddings: one for the literature dataset and one for the ESCO dataset.

The semantic similarity between each skill in the ESCO dataset and all skills in the literature dataset was measured using cosine similarity⁵. Cosine similarity quantifies how closely two vectors align in a high-dimensional space, and is defined as follows:

$$S(v_j^{esco}, v_i^{lit}) = \frac{v_j^{esco} \cdot v_i^{lit}}{\|v_j^{esco}\| \|v_i^{lit}\|} \quad (2)$$

where $v_j^{esco} \cdot v_i^{lit}$ denotes the dot product of vectors v_j^{esco} (the embedding of an ESCO skill) and v_i^{lit} (the embedding of a soft skill from the literature), and $\|v_j^{esco}\|$ and $\|v_i^{lit}\|$ are the Euclidean norms

(magnitudes) of the of the vectors. The cosine similarity score ranges from -1 to 1 , where 1 indicates identical directionality (maximum semantic similarity), 0 indicates orthogonality (no similarity), and -1 indicates opposite directionality.

For each ESCO skill s_j , cosine similarity scores were computed against all soft skills in the literature dataset. The maximum cosine similarity score $S_{max}(v_j^{esco})$ was then selected for each ESCO skill, representing the strongest semantic match:

$$S_{max}(v_j^{esco}) = \max \{S(v_j^{esco} \cdot v_1^{lit}), S(v_j^{esco} \cdot v_2^{lit}), \dots, S(v_j^{esco} \cdot v_n^{lit})\} \quad (3)$$

Thus, for each ESCO skill, the highest similarity score was identified as the best match to the set of predefined soft skills.

A threshold-based approach was employed to classify which ESCO skills could be considered soft skills. This threshold was selected to distinguish relevant soft skills from other skill types, with higher similarity scores indicating a stronger semantic relationship to soft skills. If $S_{max}(v_j^{esco}) \geq \theta$, then s_j is classified as a soft skill.

To ensure the robustness of the threshold selection, a manual review of skills near the threshold (both above and below) was conducted. This validation step ensured that the threshold provided an optimal balance between capturing relevant soft skills and excluding irrelevant ones. Skills classified as soft skills were reviewed for false positives (incorrect inclusions), and those excluded were checked for false negatives (incorrect exclusions).

This approach ensured that the model captured relevant soft skills with accuracy and minimised misclassification.

4 Results and discussion

The research led to the development of a collection of soft skills within the ESCO framework, as presented in Annex 3. This collection was derived through a matching process between the soft skills identified in the literature and those listed in ESCO, yielding several key insights. The initial phase of the study demonstrated that, out of 202 soft skills identified in the literature review, 41 skills were found to have an exact match within the ESCO framework. Of these, 15 were identified as preferred label, while 26 appeared among the alternative labels. This represents approximately 0.107% of preferred labels and 0.029% of alternative labels being identified as soft skills through exact matching, see Table 1. Although the percentage of exact matches is relatively low, these matched skills represent core competencies widely recognised in both academic research and professional frameworks as essential soft skills.

⁵ Cosine similarity scores between vector embeddings were computed using the “cosine_similarity” function from the “sklearn.metrics.pairwise” module, as supported by the “sentence-transformers” framework library in Python.

Table 1 Exact matches between literature soft skills and ESCO skills

Preferred label	Alternative labels
Systems thinking	System thinking , integrated thinking, integrate thinking, systems thought, integration of thinking
Meet commitments	Meet commitment , show conscientiousness, keep promises, meet targets, show reliability, demonstrate commitment, show responsibility, show commitment, achieve goals, demonstrate steady effort, demonstrate conscientiousness, meeting commitments, demonstrate punctuality, demonstrate reliability, work to deadlines, demonstrate loyalty
Customer service	Customer services , customer care , customer support, customer servicing, customer well being, customer satisfaction
Communication	Conversation, interaction, correspondence, communicating
Assume responsibility	Demonstrate a conscientious attitude, show responsible behaviour, be ready to take on responsibility, be responsible, showing responsibility, act responsibly, prove sense of responsibility, be ready to take responsibilities, demonstrate responsibility
Cope with uncertainty	Handle ambiguity, dealing with uncertainty, deal with uncertainties, handle uncertainty, manage uncertainty
Entrepreneurship	Business initiative, entrepreneurialism
Risk management	Risk assessment, risk administration, risk governance, hazard management
Conflict management	Conflicts management , conflict resolution, conflict supervision, conflict administration, management of conflicts
Ethics	Moral studies, the study of moral principles
Make decisions	Take decisions, choose, make decision , making decisions , decide, make choices
Listen actively	Maintain active listening, apply active listening techniques, use active listening techniques, listen actively, active listening , listen with attentive participation
Demonstrate curiosity	Display curiosity, demonstrating curiosity, show curiosity, demonstrate interest, show interest, be curious, curiosity , exhibit curiosity
Plan	Implement planning, complete planning, planning performance, plan activities, performance planning, undertake planning
Tolerate stress	Permit stress, tolerating stress , stress toleration, endure stress, abide stress, stress tolerance
Apply change management	Alter management, administer change management, applying change management, change management application, change management, revise management, implement change management, rework management, apply changing management, utilise change management , apply change management, employ change management
Demonstrate curiosity	Display curiosity, demonstrating curiosity , show curiosity, demonstrate interest, show interest, be curious, curiosity, exhibit curiosity
Approach challenges positively	Show sense of humour, keep a positive attitude, adopt a constructive approach, maintain a positive attitude
Exercise self-control	Self-control , apply self-control, exercising self-control, exercise self-control
Reflexion	Reflexion on own behaviour, mindfulness, reflexion on own behaviour, self-awareness , self-reflection
Show empathy	Demonstrate consideration, exhibit consideration, show concern for others, show courtesy, communicate an empathic response, display consideration, demonstrate concern for others, treat people fairly, show good manners, concern for others , showing empathy, be empathic
Cope with stress	Deal with pressure, perform under pressure, accept challenges, cope with pressures, work under pressure , manage pressure, handle pressure, show mental resilience, stay cool under pressure, coping with pressure
Develop strategy to solve problems	Devise strategy, come up with strategy to solve problems, set goals, set targets, design strategy to solve problems, developing strategy to solve problems, identify goals, develop strategies to solve problems, develop strategy to solve problem, create implementation strategy, adapt implementation strategy, strategic thinking , create strategy to solve problems, devise action plan, develop objectives, develop problem solving strategy, identify tasks, decide priorities
Morality	Ethics, decency, honour, integrity , morals , moral
Listen actively	Maintain active listening, apply active listening techniques, use active listening techniques, listen actively, active listening , listen with attentive participation
Conflict management	Conflicts management, conflict resolution , conflict supervision, conflict administration, management of conflicts
Tolerate stress	Permit stress, tolerating stress, stress toleration , endure stress, abide stress, stress tolerance

Skills that are exact matches for soft skills are marked in "italic bold"

The matched skills encompass a broad spectrum of essential competencies critical for both personal and professional growth. These include key communication skills, such as “active listening” and “communication”, which are essential for effective interpersonal interactions. Additionally, skills like “conflict management”, “conflict resolution”, and “customer service” highlight the importance of human interactions, both in internal teams and client-facing roles. Leadership and decision-making

abilities, such as “entrepreneurship”, “strategic thinking”, “make decisions” “plan”, and “change management”, highlights the important role of leadership in dynamic environments. Personal attributes such as “self-awareness”, “self-control”, “integrity”, and “concern for others” indicates the growing recognition of emotional intelligence as a critical professional competency. These traits, along with “stress tolerance” and the ability to “work under pressure”, underscore the need for resilience in modern

workplaces. The ability to “cope with uncertainty” and maintain a “positive attitude” reflects the increasing importance of adaptability and a growth mindset. Furthermore, the inclusion of skills such as “systems thinking”, “risk management”, “ethics”, and “morals” highlights the growing focus on holistic, ethical decision-making and responsibility in the professional sphere. The presence of “curiosity” and “maintain a positive attitude” indicates the significance of continuous learning and an optimistic approach to problem-solving and workplace challenges.

Although the exact matches between the literature-based soft skills and the ESCO framework were limited, these matches helped identify several key soft skills. However, further investigation was required to fully explore the alignment between the two sources. By examining the alignment through semantic similarity, conceptually related skills that were not captured through exact matching alone were identified. This approach provided a more comprehensive understanding of which skills within the ESCO framework could be considered soft skills, thereby expanding the scope beyond exact matches. The analysis involved examining both the similarity scores of the preferred labels and the associated alternative labels within the ESCO dataset. To enhance the robustness of the semantic analysis, exact-matched skills were retained within the NLP analysis as control elements, enabling the benchmarking of similarity scores and validation of the model’s performance. Only skills that scored above a similarity threshold of $\theta = 0.70$ were classified as soft skills. Specifically, ESCO skills were categorised as soft skills if either their preferred label or at least one of their alternative labels had a similarity score equal to or greater than 0.70. This approach ensured that conceptually related skills with high semantic similarity were not excluded

due to minor labelling differences between preferred and alternative labels. The decision to apply a 0.70 similarity use this threshold was informed by a detailed manual review of skills with scores surrounding this cut-off near the threshold, both above and below. Particular attention was given to the differences in similarity scores between preferred labels and their associated alternative labels, especially in cases where there were significant variations in scores. To further minimise the risk of false negatives, we conducted a focused audit of borderline cases within the 0.60-0.70 range, which would typically be excluded under a strict threshold-based approach. This review was structured in four intervals to assess potential conceptual alignment with soft skills definitions. In the first interval, we reviewed 209 skills scoring between 0.600 and 0.650; none of these were classified as soft skills. In the second interval, covering scores from 0.625 to 0.650 (141 skills), no additions were made either. In the third interval, which included 136 skills scoring between 0.650 and 0.675, three skills were identified as conceptually relevant and were added to the final list. Finally, in the 0.675-0.700 range, a total of 79 skills were reviewed, resulting in the inclusion of six additional soft skills. Skills initially classified as soft skills based on their similarity scores were further evaluated semantically and contextually. Following this final review, 22 skills, primarily within the 0.700 to 0.750 range, were excluded, as they did not meet the conceptual criteria for soft skills despite exceeding the similarity threshold. The manual review was conducted independently by both authors, following the same coding protocol used in the initial literature-based soft skill identification, as outlined in the methodological section. A complete list of the skills that were added and excluded through this process is provided in Table 2.

As a result, 242 concepts (preferred labels along with a total of 1,481 associated alternative labels) from the ESCO framework were categorised as soft skills. The remaining skills (13,697 preferred labels and 85,369 alternative labels), either with similarity scores below the 0.70 threshold or excluded upon manual review, were classified as non-soft skills and omitted from the final categorisation.

The distribution of soft skills within the ESCO framework indicates that a significant portion of identified skills are broadly applicable across sectors, with a concentration in cross-sector and sector-specific categories, see Fig. 1.

The analysis identified 113 cross-sector skills as soft skills, representing 46.69% of all ESCO entries labeled as soft. These competencies exhibit broad applicability across diverse professional environments. Skills such as “apply diplomatic principles”, “create solutions to problems”, and “manage feedback” illustrate essential qualities such as adaptability, effective communication, and

Table 2 Soft skills added and non-soft skills excluded after manual review of skills near the similarity threshold

Category	Skill
Included soft skill	Draw up risk assessment, adapt leadership styles in healthcare, deliver outstanding service, adapt text culturally, write stress-strain analysis reports, consult team on creative project, establish connections with young people, organise creative performance, cooperate at inter-professional level
Excluded non-soft skill	Analytical chemistry, make clinical decisions, spatial planning, literary criticism, services marketing, financial management, marketing management, quantitative analysis, implement ICT risk management, ICT communications protocols, risk financing techniques, task algorithmisation, quantitative risk analysis techniques, assessment of risks and threats, risk identification, process applications, media formats, advise on environmental risk management systems, digital media, management consulting, cognitive computing, systems theory

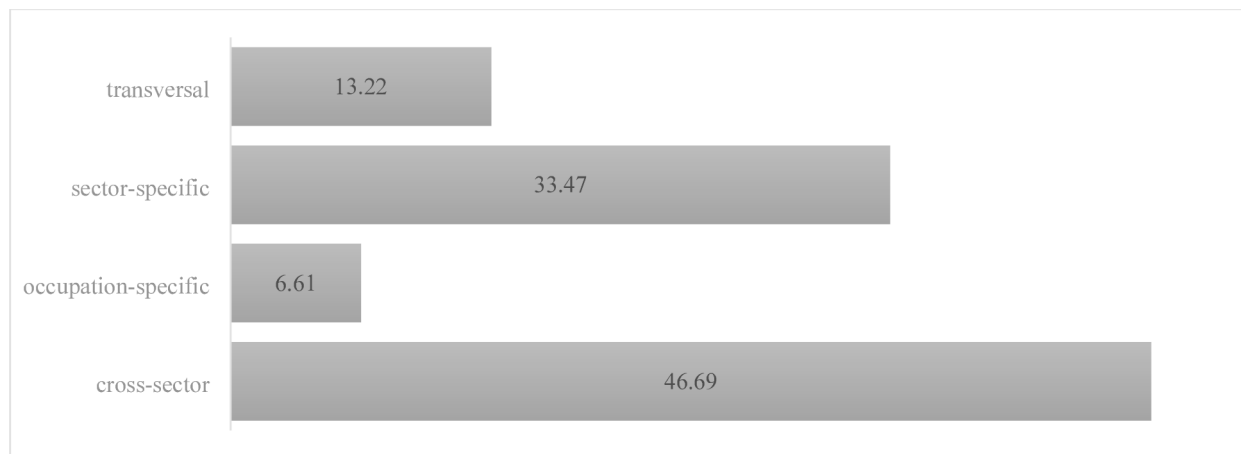


Fig. 1 Distribution of soft skills among ESCO skill by reuse level (%)

interpersonal engagement, attributes foundational to employability and valued across sectors (Ananiadou and Claro 2009; Goleman 1996). This substantial proportion reflects the high transferability of cross-sector skills, underscoring the labour market's emphasis on versatile competencies that enable individuals to navigate varied workplace environments successfully.

Conversely, 81 of the skills identified as soft were drawn from the ESCO sector-specific skills category. They account for 33.47% of the total soft skills identified and reflect competencies considered essential within particular industries. Sector-specific competencies labeled as soft skills include “enterprise risk management”, “customer insight”, and “social alliances”. These findings resonate with the literature, which suggests that certain soft skills hold specialised significance within particular sectors, providing targeted interpersonal and cognitive abilities that enhance performance within industry-specific contexts (Succi 2019).

The study also identified 32 soft skills classified under the ESCO transversal skills category, accounting for 13.22% of the total soft skills identified. These competencies are universally valued across sectors. Examples include “show initiative”, “exercise self-control”, “accept criticism and guidance”, and “meet commitments”, all of which foster resilience, collaboration, and adaptability, rendering them indispensable in diverse professional settings. Certain transversal skills, such as “show initiative” and “exercise self-control”, are particularly pertinent in leadership roles, while competencies like “accept criticism and guidance” and “work in teams” facilitate growth through feedback and cooperation. Furthermore, the analytical competency “think critically” emerged as essential for decision-making and problem-solving across varied contexts, reinforcing the literature's emphasis on critical thinking as a core professional asset (Goleman 1996). These transversal skills are consistent with the

findings of Chamorro-Premuzic et al. (2010), who highlight the role of emotional intelligence in managing interpersonal dynamics effectively within workplace settings.

In a smaller proportion, occupation-specific skills constituted 6.61% of the total matched ESCO soft skills, with 16 competencies identified as essential for specialised roles but less transferable across broader contexts. Examples of these highly specialised competencies include “systems thinking”, “patient autonomy”, and “collaborate with coaching team”, each vital within niche roles requiring advanced technical knowledge. This limited representation aligns with Clarke (2018) findings on the need for specialised soft skills in roles where technical expertise and domain-specific knowledge take precedence over general adaptability.

Across categories, communication and problem-solving emerged as foundational skills that consistently appeared in various ESCO categories. Communication facilitates effective interpersonal interactions, conflict resolution, and team dynamics, critical for both internal collaboration and client-facing functions. Similarly, problem-solving fosters critical thinking and innovation, equipping individuals to address complex challenges, a crucial competency in fast-paced, evolving professional environments. The high prevalence of these competencies across categories underscores their centrality in fostering productivity and cohesion, aligning with studies by Davis-Blake et al. (2002) and Robles (2012) that emphasise these competencies as integral to both personal and professional success. Further supporting this alignment, transversal competencies such as “adapt to change” and “show initiative” highlight the importance of behavioral adaptability in dynamic work environments, echoing Heckman and Kautz (2012) findings on dispositional traits as core to soft skills. Skills associated to emotional intelligence highlighted by Goleman (1996), were also identified as soft skills, with labels such

as “show empathy” and “exercise self-control”. These findings reinforce Chamorro-Premuzic et al. (2010) assertion regarding the critical role of emotional intelligence in addressing both individual and relational challenges in the workplace,

The categorisation of soft skills within the ESCO framework reveals a varied distribution, with a predominance of cross-sector (46.69%) and sector-specific (33.47%) competencies. While some soft skills are clearly context-dependent, a significant portion reflects broad, transferable applicability across sectors. This structured classification strengthens ESCO’s alignment with labour market requirements, supporting a blend of foundational and specialised soft skills essential for employability and professional development across a wide array of sectors.

5 Conclusions

This research represents a substantial advancement in standardising soft skills terminology and integrating them into established frameworks like ESCO, with significant implications for labour markets, education, and policymaking. It contributes to the ongoing dialogue highlighted by scholars such as Heckman and Kautz (2012) and Cinque and Ciappei (2014) on the necessity of consistent soft skill classification. A key outcome of this study is the development of a comprehensive collection of soft skills based on the ESCO taxonomy, providing a structured foundation for soft skills categorisation. The findings confirm the widespread relevance of soft skills, both cross-sectoral and sector-specific, underscoring their critical role in today’s dynamic workforce. This corroborates earlier research on the transferability and adaptability of soft skills in various contexts (Clarke 2018; Succi 2019). As automation, artificial intelligence, and other advanced technologies continue to reshape occupational landscapes, the distinct categorisation of soft skills within the ESCO framework highlights their growing importance alongside the technical skills that have traditionally dominated labour market models.

The refined classification of soft skills offers practical applications for researchers, educators, policymakers, and HR professionals, facilitating its integration into diverse contexts. For researchers, the classification provides a structured foundation for designing studies that investigate the role and impact of soft skills across various sectors. Educators can utilise this taxonomy to align curricula with labor market demands, emphasising key soft skills such as “communication” and “problem-solving” to enhance employability. Policymakers can apply the classification to develop targeted workforce strategies, address skills gaps, and promote labour mobility. Similarly, HR professionals can leverage the classification to design competency frameworks and assessment tools for recruitment and professional development. Moreover,

developers of digital tools (e.g., job-matching or skill mapping platforms) can integrate the classification into NLP based systems to improve automated alignment between job descriptions, candidate profiles and training contents.

In addition, introducing a distinct “soft skills” label—similar to ESCO’s existing “green” and “digital” skill tags would enhance visibility, support standardisation, and promote targeted development initiatives.

Several recommendations emerge from this study to further refine the ESCO framework. First, expanding the emotional intelligence (EI) skill within ESCO would align with the growing importance of EI in professional contexts. While ESCO includes competencies like “show empathy” and “exercise self-control”, the literature highlights a broader range of EI-related skills, such as managing interpersonal relationships and accurately perceiving others’ emotions. Adding detailed labels for these competencies would enrich ESCO’s EI representation and better meet workforce demands. As proposed by Fareri et al. (2021), automated identification of such nuanced skills can support HR professionals and educators in developing more accurate soft skill assessments, ensuring that vital interpersonal competencies are both identifiable and trainable across sectors. Another area for improvement involves the explicit contextualisation of soft skills across different sectors. ESCO could benefit from a clearer differentiation of how soft skills, such as communication and teamwork, are operationalised in distinct industries. For example, effective communication in healthcare requires empathy and patient-centered clarity, while in customer service, it involves responsiveness and problem resolution. Refining ESCO to reflect these contextual variations would enhance its applicability and precision. Furthermore, while ESCO include references to creativity, the increasing emphasis on innovation suggests a more nuanced approach. The literature underscores creativity’s importance not only in traditional arts but also in problem-solving and strategic roles across diverse sectors. ESCO could expand labels related to creative thinking, particularly within transversal and cross-sector skills, to capture the shift toward innovation-driven roles. Adaptability and the willingness to learn have also emerged as essential transversal competencies due to rapid industry evolution. As employees are required to continuously update their knowledge and skills, ESCO could incorporate more specific soft skills related to lifelong learning, self-assessment, and proactive self-development. This would ensure that the framework addresses the growing need for individuals to adapt to ongoing professional growth. Finally, while the study confirmed a semantic alignment between the soft skills identified in the literature and skills within ESCO, more refinement in terminology is needed. Terms like

“problem-solving” and “critical thinking” have broad applications but vary in sectoral nuances. ESCO could consider adopting more precise language that captures both the overarching relevance of these skills and their specific sectoral manifestations.

While this study contributes to the clarity of soft skills within the ESCO framework, certain limitations should be acknowledged. Difficult access to relevant data constrained the literature review, as many articles were not open-access. Additionally, while NLP methods provided valuable insights, they encountered challenges in fully capturing contextual nuances due to the reliance on pre-trained models rather than a custom-trained model specific to soft skill analysis. This reliance on generalised data sources limited the model's precision, particularly in achieving high contextual relevance. Although pre-trained SBERT embeddings performed well, a custom-trained SBERT model tailored specifically for soft skills would likely enhance precision and alignment. Another methodological challenge involved establishing an optimal semantic similarity threshold to classify skills accurately; while the selected threshold yielded meaningful classifications, it introduced some subjectivity, potentially impacting the consistency of soft skill identification across diverse contexts. Moreover, the inherent subjectivity in defining soft skills across various disciplines, industries, and cultural contexts adds complexity to efforts toward standardised classification. Future research could refine this classification process by integrating more comprehensive soft skills datasets and by further exploring advanced customised NLP models that can capture contextual nuances with higher accuracy.

Despite these limitations, the findings presented in this study offer meaningful progress in classifying soft skills within a structured framework. The proposed taxonomy enhances theoretical understanding and offers practical tools for stakeholders seeking to align education, workforce development, and policy with the dynamic demands of the modern labour market. Continued refinement and adoption of such classifications will be essential to ensure that soft skills remain visible, measurable, and actionable in a labour market shaped by technological innovation and global integration.

Supplementary Information

The online version contains supplementary material available at <https://doi.org/10.1186/s12651-025-00409-x>.

Supplementary Material 1

Supplementary Material 2

Supplementary Material 3

Author contributions

A.G. participated in data collection, conducted the literature review, drafted the main manuscript text, assisted in interpreting the results, and contributed to the revision process. C.P. participated in data processing, performed the analysis, interpreted the results, and contributed to writing and revising the document.

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Data availability

The dataset of soft skills extracted from the literature (Annex 1) and the soft skills collection generated through the matching process (Annex 2) are provided as electronic supplementary files.

Declarations

Ethics approval

This article does not contain any studies with human participants performed by any of the authors.

Informed consent

This article does not involve any studies conducted with human participants by the authors; therefore, obtaining consent was not required.

Competing interests

The authors declare that they have no competing interests.

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